

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |

FEB  
2021

WK 04 - 018-347

# 4-Theory-HW

2021

MONDAY  
JANUARY

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①

CT = H L O T Q I H J P W X Q C M A G

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \quad \text{PT} = ?$$

$$|K| = 21 - 10 = 11$$

$$|K|^{-1} = 11^{-1} \pmod{26} = 19$$

$$[\det K]^{-1} \text{adj}[K]$$

$$= 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$\text{HL} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{matrix} T \\ O \end{matrix}$$

$$\text{PT} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$\text{QI} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{matrix} E \\ C \end{matrix}$$

$$\text{HJ} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} \pmod{26} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ F \end{matrix}$$

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TUESDAY  
JANUARY

2021

|          | S  | M  | T  | W  | T  | F  | S  |
|----------|----|----|----|----|----|----|----|
| DEC 2020 | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|          | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
|          | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|          | 27 | 28 | 29 | 30 | 31 |    |    |

$$P \times \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 2 \end{bmatrix} = \begin{bmatrix} T \\ A \end{bmatrix}$$

$$WQ \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 290 \\ 271 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} \begin{matrix} E \\ S \end{matrix}$$

$$CM \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 174 \\ 88 \end{bmatrix} \pmod{26} = \begin{bmatrix} 16 \\ 6 \end{bmatrix} \begin{matrix} S \\ A \end{matrix}$$

$$AG \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} \begin{matrix} G \\ H \end{matrix}$$

PT = TOP SECRET MESSAGE

② Hill Cipher

$$K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

$$CT = RXAOPY \quad PT = ?$$

$$|K| = 24 \times 14 - 5 \times 19 = 241 \pmod{26} = 7$$

$$7^{-1} = 15$$

$$15 \begin{bmatrix} 14 & 19 \\ -5 & 24 \end{bmatrix} \pmod{26} = \begin{bmatrix} 210 & -285 \\ -75 & 320 \end{bmatrix} \pmod{26} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$RX \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 547 \\ 587 \end{bmatrix} \pmod{26} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \begin{matrix} R \\ L \end{matrix}$$

We must believe that we are gifted for something, and that this thing, at whatever cost, must be attained. -Marie Curie

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
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| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |

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WK 04 • 020-345

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$$\begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \text{mod } 26 \rightarrow \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 21 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 531 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 4 \\ 19 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

PT = FLOWER

### ③ Relationship b/w

#### i) Substitution & Confusion

\* Substitution replaces one character with another character

HELLO → KRIIR

④ Substitution creates confusion by hiding the relationship b/w original plain text character and cipher text.

\* makes it difficult to understand how the key and plain text are related to the cipher text.

## 2) Transposition and Diffusion:

Ⓐ Transposition changes the position of character, with respect to changing the characters themselves.  
TRANS - SANTR

\* Transposition creates diffusion by spreading the influence of plaintext character across diff. position in CT.

Ⓑ Diffusion spreads info of the plaintext throughout the ciphertext making patterns harder to recognise.

| Cipher                 | Does it change the char / symbol value | Change Char position | Confusion / Diffusion | Justification                                                                                                      |
|------------------------|----------------------------------------|----------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Shift Ciphers          | Yes                                    | No                   | Confusion             | Replaces each char with a diff character based on addition key, securing the PT without moving characters location |
| multiplication ciphers | Yes                                    | No                   | Confusion             | Replace each character with another character using multiplication                                                 |

|    |                       |     |     |           |                                                                                                                                        |
|----|-----------------------|-----|-----|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Affine Cipher         | yes | no  | Confusion | Combines both shift & multiplication to establish char entirely focusing on altering the symbols                                       |
| 10 | Transposition Ciphers | no  | yes | Diffusion | Keeps all <u>orig</u> chars but scatters them to different positions spreading the <del>positions</del> pattern address to cipher text |
| 11 |                       |     |     |           |                                                                                                                                        |
| 12 |                       |     |     |           |                                                                                                                                        |
| 1  |                       |     |     |           |                                                                                                                                        |

## Transposition :

### Encryptions

Encryptions = The arrow is pointing north direction

- key = 4
- keyword = ATLAS
- double :  $k_1 = \text{BREAK}$   
 $k_2 = \text{LOG}$

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TUESDAY  
JANUARY

2021

|          | S  | M  | T  | W  | T  | F  | S  |
|----------|----|----|----|----|----|----|----|
| DEC 2020 | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|          | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
|          | 20 | 21 | 22 | 23 | 24 | 25 | 26 |
|          | 27 | 28 | 29 | 30 | 31 |    |    |

Decryption:

TIDIE MRNME REO40 SANOW RSLTP EEED0  
SSNVU DHTUD BIEN9 GODAE PEIXD ELMSX

→ Simple:  $K = 10$   $R = \frac{60}{10} = 6$

|   |              |   |   |   |   |   |   |   |   |   |
|---|--------------|---|---|---|---|---|---|---|---|---|
| 1 | <del>T</del> | M | E | N | T | O | A | I | D | X |
| 2 | <del>I</del> | R | O | O | P | S | H | E | A | D |
| 3 | <del>D</del> | N | G | W | E | S | T | N | E | E |
| 4 | <del>I</del> | M | O | R | E | S | U | P | P | L |
| 5 | <del>E</del> | E | S | S | E | N | D | G | E | N |
| 6 | <del>M</del> | R | A | L | D | U | B | O | I | S |

TRU

|   |              |   |   |   |   |   |   |   |   |   |
|---|--------------|---|---|---|---|---|---|---|---|---|
| 1 | T            | R | O | O | P | S | H | E | A | D |
| 2 | I            | N | G | W | E | S | T | N | E | E |
| 3 | D            | M | O | R | E | S | U | P | P | L |
| 4 | <del>T</del> | E | S | S | E | N | D | G | E | N |
| 5 | E            | R | A | L | D | U | B | O | I | S |
| 6 | M            | E | N | T | O | A | I | D | X | X |

TROOPS HEADING WEST NEED MORE SUPPLIES  
SEND GENERAL DUBOIS MENT TO AID.

| M  | T  | W  | T  | F  | S  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |

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i) Keyword = SECRET

1 L E D T N H P T R S O I I T M A X S A X E R X T  
A N O N I S W F O T X

$$R = \frac{36}{6} = 6$$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | T | H | I | S | T | S |
| 2 | E | P | I | A | A | W |
| 3 | E | T | T | X | N | F |
| 4 | P | R | M | E | O | O |
| 5 | T | S | A | R | N | T |
| 6 | N | O | X | X | I | X |

PT = THIS IS A PLAIN TEXT FOR  
DEMONSTRATION

ii) KF PLAIN

CT = KEY BAX ARA OTX H4P

$$R = 15/5 = 3$$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | H | A | V | E | A |
| 2 | G | R | E | A | T |
| 3 | D | A | Y | X | X |

PT = HAVE A GREAT DAY

Our goal is to make the best devices in the world, not to be the biggest.

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THURSDAY  
JANUARY

2021

# Permutation

Find Inverse Permutation

① 4 3 1 5 2

|  |   |   |   |   |   |         |
|--|---|---|---|---|---|---------|
|  | 1 | 2 | 3 | 4 | 5 | → index |
|  | 4 | 3 | 1 | 5 | 2 |         |

Inverse : 3 5 2 1 4

② 1 2 3 4 5 6 7  
7 4 1 3 5 2 6

Inverse : 3 6 4 2 5 7 1

③ 1 2 3 4  
4 1 3 2

Inverse : 2 4 3 1

④ 1 2 3 4 5  
3 5 1 4 2

Inverse : 3 5 1 4 2